

Healthcare AI/ML Case Study

Predictive Diagnosis of Diabetes & Reinforcement-Learning-Based Personalised Treatment Recommendation

Business Context

A healthcare company has collected patient records — age, blood pressure, cholesterol, glucose level, BMI, and family history — with the objective of building a predictive model for the early diagnosis of diabetes. In addition, the company wants to train an AI agent that recommends personalised treatment actions (Diet / Exercise / Medication / Monitor) based on how a patient's health state transitions over time.

Role: You are hired as an AI/ML Engineer and are required to address the four tasks below, covering inductive learning & data preparation, decision-tree-based diagnosis, statistical risk stratification, and reinforcement-learning-based treatment recommendation.

Task 1 — Data Preparation & Inductive Learning

- (a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2 — Decision Tree for Diagnosis

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them — justify clinically.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs. post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3 — Statistical Learning for Risk Stratification

- (a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4 — Reinforcement Learning for Treatment Recommendation

- (a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
- (b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

Deliverables: Problem Statement (this document), Solution Report, Python Implementation (.py), Output Visualisations (.png), Consolidated Project Report, and README.